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PHILRICE R&D HIGHLIGHTS



Crop Biotechnology Center



Philippine Rice Research Institute
Central Experiment Station
Maligaya, Science City of Muñoz, 3119 Nueva Ecija

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Crop Biotechnology Center

CENTER DIRECTOR: ROEL R. SURALTA

EXECUTIVE SUMMARY

The Research and Development (R&D) operations of the newly constructed and improved Crop Biotechnology Center (CBC) are well underway, aiming to develop new crop varieties with higher yields, reduced input requirements, enhanced climate resilient, and improved nutritional value. The center hopes to achieve these goals by employing advanced biotech solutions to contribute to the advancement of crop science and technology and to the enhancement of value, availability, and utilization of diversified farming products for better quality, safety, health, nutrition, and income of farmers.

The CBC, through its research activities, has published 13 scientific publications in local and international journals: Nature, Proceedings of the National Academy of Sciences (PNAS) of the United States of America, European Journal of Agronomy, Plants, Philippine Journal of Science, Rice-based Biosystems Journal, and BPI- NCT Manual.

New Breeding Techniques (NBTs) were employed in different studies to enhance crop traits and develop novel breeding lines for various traits: rice tungro spherical virus (RTSV) resistance, bacterial leaf blight (BLB) resistance, and amylose content. Genome sequences of mutant lines GXB and NSIC_Rc222SM2 alongside their wild-type counterparts, Gal_Ong and NSIC Rc 222 were aligned with the reference genome. Biosafety permits were granted for field trials of nutritionally improved rice with high-iron (Fe) and zinc (Zn) rice varieties and were conducted in Science City of Muñoz, Nueva Ecija and Batac City, Ilocos Norte. Golden Rice Event GR2E with PSB Rc 82 background was registered in the National Seed Industry Council (NSIC) as NSIC 2022 Rc 682GR2E or Malusog 1.

Genetic engineering is also an important biotech tool that the center has employed in the developing recombinant bacteria capable of lignin-degrading enzymes and rice tungro coat proteins. To aid in molecular marker-assisted rice breeding, the center has identified and validated quantitative trait loci (QTLs) associated with root plasticity, functional stay-green, and rice tungro virus (RTV) resistance traits. Thirteen (13) QTLs are associated with lateral root plasticity under water stresses, four for nodal root plasticity, seven for functional stay-green traits, and two for rice pest and disease resistance.

In compliance with DA Administrative Order 26, Series of 2021, the center reviewed the 2018 Crop Biotechnology Research, Development, and Extension (RDE) Agenda and developed a new Research for Development and Extension (R4DE) Agenda. New crop biotechnology knowledge products (KPs) were also produced (pamphlets, news articles, and briefers), and launched its official social media page to support its information, education, and communication (IEC) efforts. Capacity- building initiatives were also provided by the center such as Molecular Biology Techniques Training and thesis support for graduate students from Benguet State University, Nueva Ecija University Science and Technology, UP Los Baños, Central Luzon State University, and Nagoya University-Japan.

CBC-212:

Genetically Engineered Bacteria with Biomass-Degrading and Fermenting Activities for Bioethanol Production from Rice Straw

Reynante L. Ordonio and Jayvee A. Garcia

The development of genetically engineered microorganisms for enhanced rice production and industrial uses—specifically, biomass-degrading bacteria and bacteria expressing fusion protein for diagnostics purposes — is one of PhilRice's innovations toward attaining a competitive rice economy through the use of biotechnology for food production, fuel, waste management, and value-added products. Work toward the development of a bacterium with a lignin-degrading activity began with the identification of genes responsible for lignin-degradation including their gene sequences and source microorganisms, using annotated gene sequences available in the GenBank database on the National Center for Biotechnology Information (NCBI) website. Expression vector containing the top candidate gene coding for lignin-degrading protein was synthesized using the gene synthesis services of SBS Genetech Co., Ltd. The candidate Lip1 gene (1104 bp) was designed for synthesis and insertion into pUC19, resulting in a plasmid-gene construct of 3,871 bp which was then transformed into *E.coli*. The gene synthesis of Lip1 with T7 was done but the orientation in the resulting vector from the manufacturer was not as expected. Nevertheless, this was used as a deoxyribonucleic acid (DNA) template for the correct ligation of the insert into the vector pUC19. Insertion into pUC19 was successful in the correct orientation and initial expression and activity of the LiP1 gene were tested using the lignin-degradation assay described by Bandounas et al., 2011.

A biosafety proposal was submitted to Department of Science and Technology–Biosafety Committee (DOST-BC) and received approval, with endorsement by the PhilRice CES Institutional Biosafety Committee (IBC); one pUC19 construct (ligand) produced with Lip1 gene from *Trametes versicolor* mushroom; and one transformed *E. coli* strain is under testing for lignin-degrading ability with Lip1 gene from *T. versicolor* mushroom.

RTF-019-283

Nutritionally Enhanced Rice: Finishing and Delivering GoldenRice and High-Iron and Zinc Rice Varieties

COMPONENT 1.1. SUPPLY VARIETAL IMPROVEMENT

**Reynante L. Ordonio, Ronalyn T. Miranda, Luilene A. Miranda, Trinidad C. Fernando,
Mark Philip B. Castillo, Marlon C. Garcillano, and Rhona Jane B. Fabros**

The Golden Rice Program comprises five project components: supply; demand; governance; planning, monitoring, and evaluation; and program management. The DA-CBC conducts varietal improvement under the supply component, which aims to achieve two key objectives: first by introgression of the Golden Rice GR2E event into popular high-yielding rice varieties; and second by establishing guidelines for seed certification and varietal registration. In 2022, the varietal improvement team established 21 advanced lines in backgrounds such as NSIC Rc 402, Rc 160, Rc 300, Rc 238, Rc 358, and Rc 298 while also successfully registering the product with PSB Rc 82 background with the National Seed Industry Council (NSIC) as NSIC 2022 Rc682GR2E or Malusog 1 on April 17, 2022. This notable achievement serves as evidence for the continuous deployment and commercialization of the Golden Rice/Malusog 1.

Four batches of PSB Rc 682 (two at FutureRice Farm, one in a screenhouse, and one at CBC) were established, harvesting more than 100kg of nucleus seeds based on established stewardship/seed production protocols; generated BC5F4 lines of NSIC Rc 300-GR2E (4), Rc 402-GR2E (6), Rc 358-GR2E (3), and Rc 298-GR2E (5), and BC5F1 of Rc 216-GR2E (10), Rc 308-GR2E (10), and Rc 160-GR2E (10) with introgressed GR trait; one line was registered as a variety (NSIC 2022

Rc 682GR2E), with 50kg of nucleus seeds available based on existing NSIC guidelines; completion report submitted and certificate of completion secured for the project with DOST-BC Ref No. 2020-0325 titled "Contained Use for the Production of Grain Samples for Grain Quality and Sensory Evaluation of Golden Rice event GR2E at PhilRice CES"; established nucleus seed production of PSB Rc 82 GR2E and harvested 12.8kg; genotyped and phenotyped by dehulling 1,040 individual plants resulting in 8.8kg of ultra-pure seeds; and advanced lines of NSIC Rc 300-GR2E, Rc 402-GR2E, Rc 358-GR2E, and Rc 298-GR2E, Rc 216-GR2E, Rc 308-GR2E, and Rc 160-GR2E were selected for preliminary yield trial under field conditions for identification of the two best lines for distinctness, uniformity, and stability (DUS) testing in preparation for varietal registration application (Uniform lines introgressed with GR traits showing pure yellow grains and homozygous to the GR locus were selected).

RTF-019-346:

Nutritionally Enhanced Rice: Finishing and Delivering Golden Rice and High-iron and Zinc Rice Varieties

High-iron and Zinc Rice Project (HIZR)

Reynante L. Ordonio, Ronalyn T. Miranda, Luilene A. Miranda, Trinidad C. Fernando, Marlon C. Garcillano, and Rhona Jane B. Fabros

The project is a collaborative effort between PhilRice and IRRI, which was under the Healthier Rice Program. It aims to develop a Philippine rice variety with high Iron (Fe) and Zinc (Zn) in the polished grains. On May 27, 2022, the project submitted two regulatory applications for field trials for the two high-iron and high-zinc rice (HIZR) events IRS1030-031 and IRS1030-039 to Department of Agriculture-Bureau of Plant Industry (DA-BPI) under the JDC no. 1 series of 2021.

The field trial aims to: (1) confirm that genetically engineered HIZR is substantially equivalent to conventional non-transgenic rice under typical production conditions; and (2) collect grain yield and other agronomic (phenotypic) measurements from field-grown plants at two locations (one primary site and one backup site). This trial will determine whether there are any unexpected, unintended effects of the genetic modification process on the normal growth and development of HIZR. In addition, the field trial will be used to generate plant material (e.g., paddy rice and straw) and derived products (e.g., milled rice

and bran) for chemical analyses. Also, some of the harvested seeds may be used if another field trial is needed. On October 20, 2022, DA-BPI issued the biosafety permits to conduct the said trials in the Science City of Muñoz (SCM), Nueva Ecija and Batac City, Ilocos Norte. In December 2022, field trials were established at both sites. Once the trials are completed, a regulatory application for commercial propagation will be submitted to DA-BPI. In the coming years, our goal is to stack HIZR with GR, which would greatly enhance nutrient supply and lead to a significant public health impact.

Two field trial proposals for HIZR events IRS1030-031 and IRS1030-039 were submitted to DA-BPI with endorsement from the PhilRice IBC; public information sheets were posted in the cities of Batac and Muñoz in preparation for the HIZR trials; an annual/progress report for project DOST-BC Ref. No. 2021-0332—covering one season of the HIZR confined test (HIZR CT) at PhilRice Batac—was submitted with endorsement from its IBCs; and four biosafety permits for the two HIZR events at PhilRice CES and Batac, with permits for field trial Nos. 22-001a-FT and 22-002a-FT, respectively, were secured.

RTF-022-326

Improving Popular Released Rice Varieties Through Gene-Editing (DA-Biotech)

Reynante L. Ordonio, Trinidad C. Fernando, Mark Philip B. Castillo, Marlon C. Garcillano, Rhona Jane B. Fabros, Luilene A. Miranda, and Ronalyn T. Miranda

Rice breeders now have a precise and faster alternative method of modifying current popular varieties using modern molecular biotechnology such as CRISPR/Cas9 or Cas12 systems. In this project, we used rice cultivars recommended by the DA's Rice Competitiveness Enhancement Fund Seed Program. The first stage was to find potential recipients, which were popular indica cultivars that respond well to tissue culture to produce calli, and explore the use of the floral dip method to produce transformed seeds. Simultaneously, appropriate constructs for CRISPR-Cas gene-editing were designed, involving the following traits: resistance to rice tungro spherical virus (RTSV), resistance to bacterial leaf blight (BLB), improved aroma, and lower amylose content than the original.

Out of 44 NSIC varieties evaluated, we selected five high yielders (NSIC Rc 160, PSB Rc 82, Rc 512, Rc 430, and Rc 426) that were receptive to tissue culture (regeneration), and these were mass-produced for *Agrobacterium* inoculation. A floral dip experiment with *Agrobacterium* was also performed, generating one

putative transformant that grew on MS media containing hygromycin. However, when its R2 generation was phenotyped for hygromycin resistance, the floral dip was found to be ineffective; hence, floral dips with different panicle cuts for emasculation were performed in pre- and post-anthesis panicles. Meanwhile, 15 primer/oligonucleotide pairs for the guide RNA component of CRISPR-Cas9 vectors for gene-editing have been designed: five for BLB resistance (OsSWEET 11, 13, and 14 genes), three for tungro virus resistance (tsv1 gene), and seven for amylose content (Waxy gene). Vector construction using the pRGEB32 plasmid for the OsSWEET 13 and 14, OsTSV1, and OsWAXY genes is ongoing.

One gene-editing construct (ligand) was produced for each trait (amylose biosynthesis, tungro, and BLB resistance); one floral dip protocol was optimized for producing one gene-edited seed per panicle; and one gene-edited line was produced at the callus stage (for further confirmation) for each trait (amylose biosynthesis, and tungro and BLB resistance).

RTF-022-249:

Development of Provitamin A-enriched and Disease-resistant Rice Lines Through Marker-assisted Breeding (DA Biotech)

Reynante L. Ordonio, Ronalyn T. Miranda, Luilene A. Miranda, Trinidad C. Fernando, Mark Philip B. Castillo, and Rhona Jane B. Fabros

Golden Rice (GR), or Malusog 1 is the first-ever genetically modified (GM) rice with grain beta-carotene that has been commercially approved and registered as a local variety. Like any other rice variety, GR is exposed to various biotic stresses, so having pest and disease resistance becomes a necessary trait for the plant to sustain its maximum yield. To create more nutritional impact on consumers, it is important to use other well-adapted elite genetic backgrounds aside from PSB Rc82. To this end, we created a pipeline of GR advanced backcross inbred lines (BC4F2-BC5F3) that preserve the yield and grain quality characteristics of their respective recurrent parents while having a degree of resistance to tungro and bacterial leaf blight (BLB), along with meaningful levels of beta-carotene in the grains.

Here, we developed “all-in-one” lines derived from four popular varieties, namely: NSIC Rc 160, Rc 216, Rc 238, and Rc 300. The lines have yellow grains with significant levels of total carotenoids. They also had intermediate and resistant reactions to induced BLB and RTV disease screening under greenhouse

conditions. The 17 advanced lines have at least one resistance gene each for tungro (tsv1, glh14) and BLB (Xa4, xa5, Xa7, Xa21). These newly developed 3-in-1 GR lines will be field-tested and ultimately registered with the NSIC to provide farmers with improved GR varieties that not only potentially improve vitamin A nutrition but also help ensure farmers' yields by way of the introduced resistance to tungro and BLB.

The report on 10 advanced lines for NSIC Rc 300 at BC5F5, 15 advanced lines (five each per background) for NSIC Rc 238, Rc 216 at BC4F4 and BC4F3, and 5 advanced lines from NSIC Rc 160 at BC4F3 with resistance and GR traits has been submitted to the DA-Biotech.

RTF 004-342

Molecular Mechanisms of Root System Formation for Genetic Improvement of Rice Adapted to Water Stress Conditions

Nonawin L. Agustin, Roel R. Suralta, Jonathan M. Niones, and Rebecca I. Santos

Phenotypic root plasticity—particularly plasticity in L-type (long, thick, and highly-branched) lateral root (LR) development—is a crucial trait for stable growth, adaptability, and productivity of rice under changing global climate conditions. Despite its significance to water stress, the genetics and mechanisms controlling this trait remain largely unknown, limiting its potential for breeding and biotechnological applications. The project aims to uncover the genetic and epigenetic regulation associated with root plasticity under fluctuating soil moisture conditions (i.e., drought and rewatering). Using selected genetic materials, particularly mutation genes promoting root system formation. We have evaluated the performance of the *weg1* mutant under water stress conditions using the pail setup; introgressed the mutation gene in widely cultivated and high-yielding Philippine rice varieties; and screened rice lines with improved root system by using an efficient root screening system.

Additionally, LR-related genes were initially examined for gene expression in 10 rice varieties and an initial QTL associated with L-type LR plasticity under rewatering conditions was found on Chromosome 1. Genotyping using additional markers is ongoing and QTL detection will be validated under rewatering. This increased understanding of the mechanisms of the trait will be useful in breeding crops to allow improved drought avoidance with higher root plasticity.

Results showed that the *weg1* mutant had greater root growth than wildtype under continuously waterlogged (control) conditions. Such a highly developed root system in the mutant was also observed under drought-induced and fluctuating moisture stresses while its shoot growth was greater than the wildtype. This suggests that *weg1* mutation gene could be a novel breeding material for drought stress avoidance, leading us to its introgression to popular and high-yielding varieties to improve their root architecture. The target backgrounds above do not possess the *weg1* mutation gene.

For efficient screening of lines targeting a highly branched root system, a combined hydroponics setup and a modified rapid generation advancement (RGA) technique was optimized, allowing 60-70% savings in time, cost, and labor both in laboratory and screenhouse for every filial generation advancement. A dCAPS marker specific to the *weg1* mutation gene designed and optimized for genotyping. A total of 597,380 and 634 F3 plants from Rc 160, Rc 402, and Rc 480 backgrounds were developed and advanced; corresponding BC1F1 are also being advanced.